

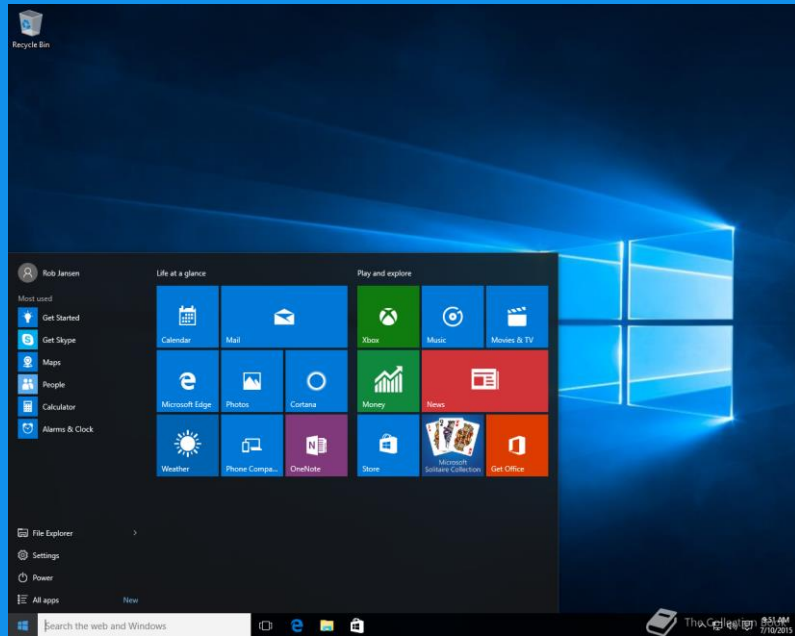
Metodi Computazionali per la Fisica

Linux e Uso del Terminale

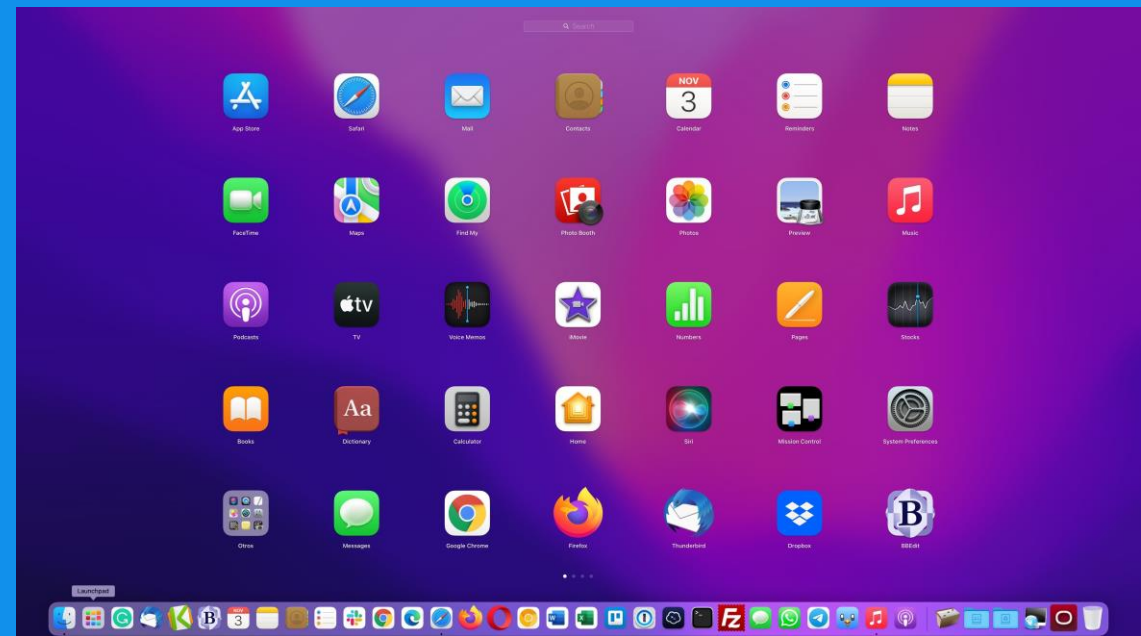
S. Germani – stefano.germani@unipg.it

Sistemi Operativi

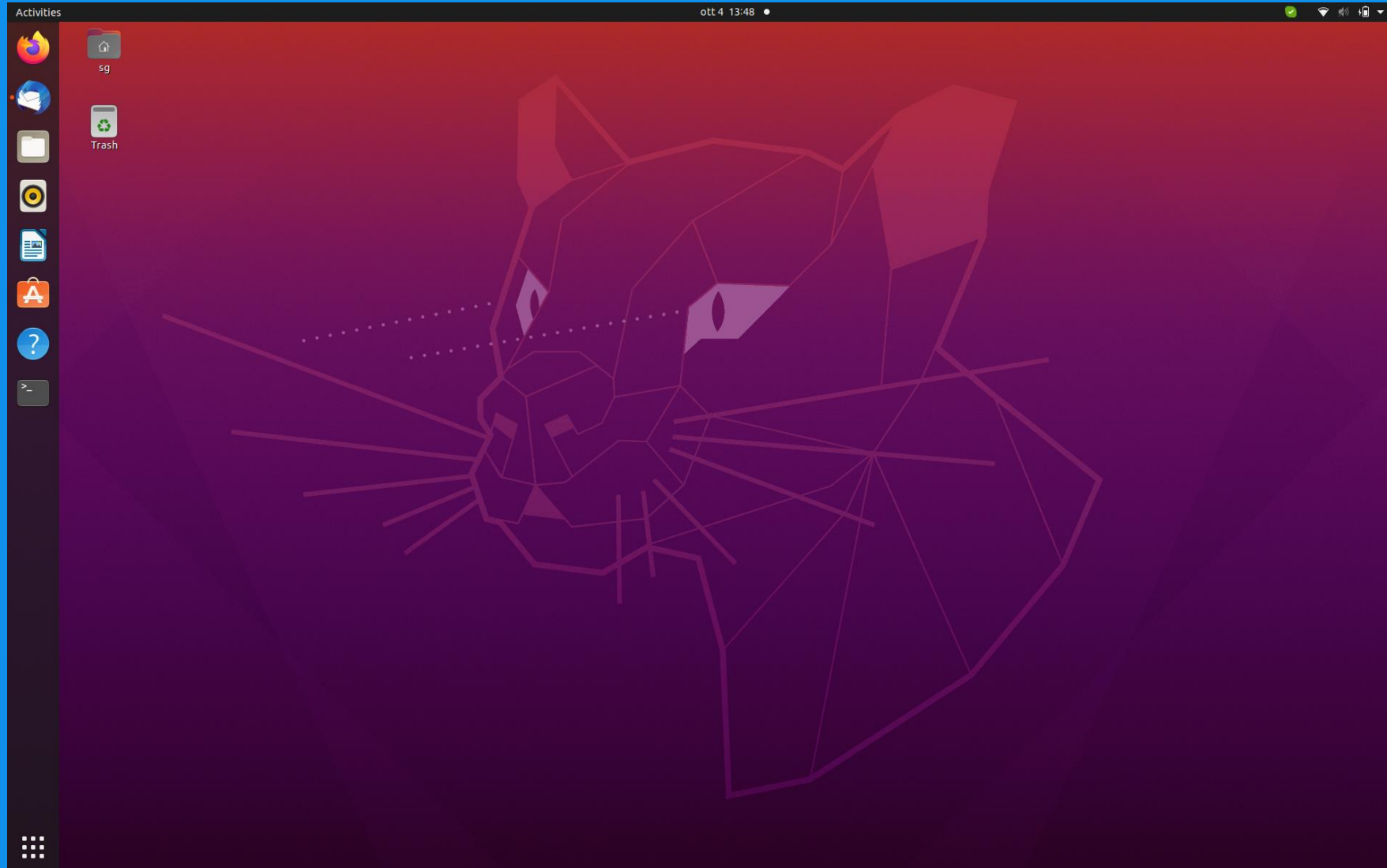
MS Windows



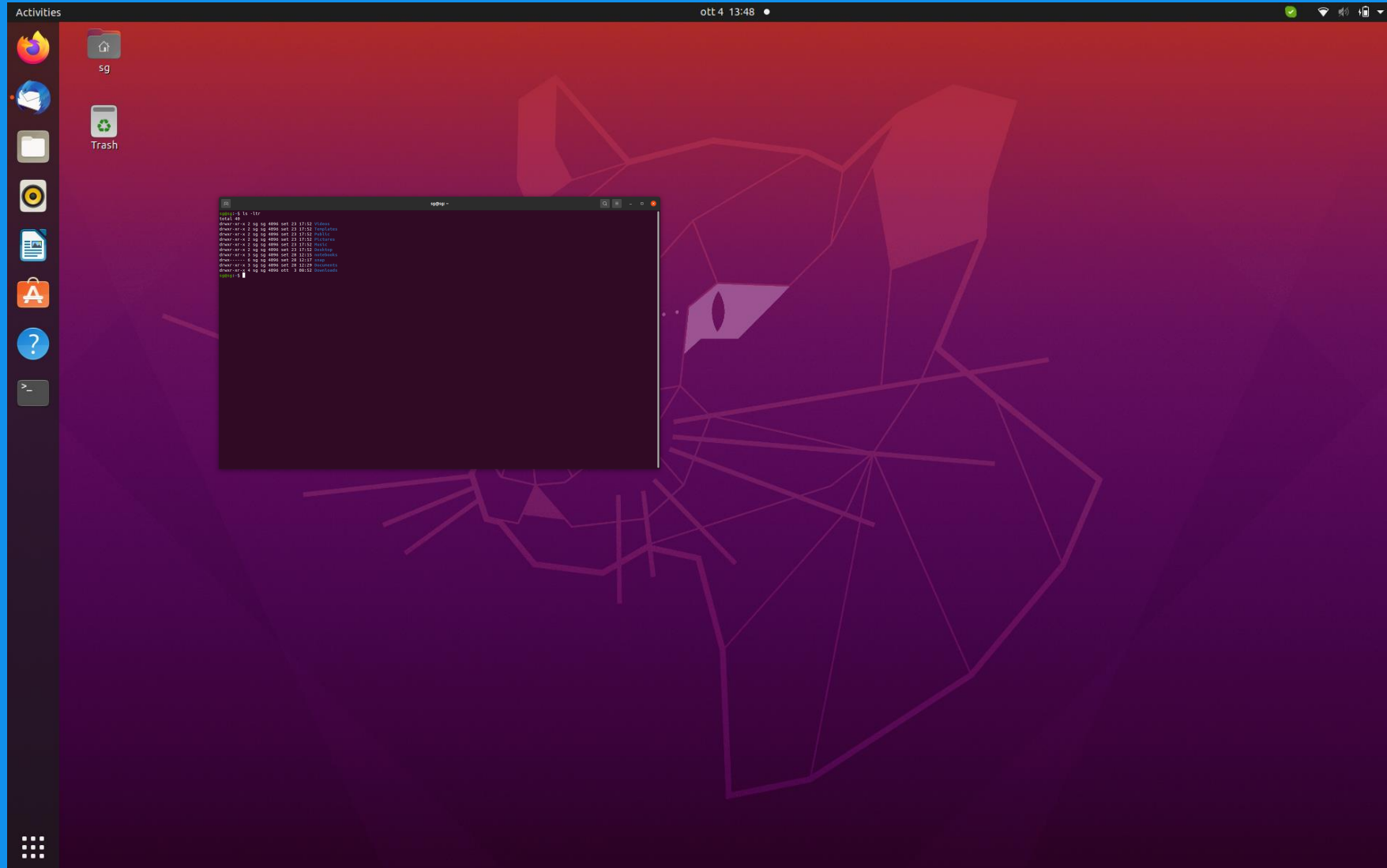
Mac OS



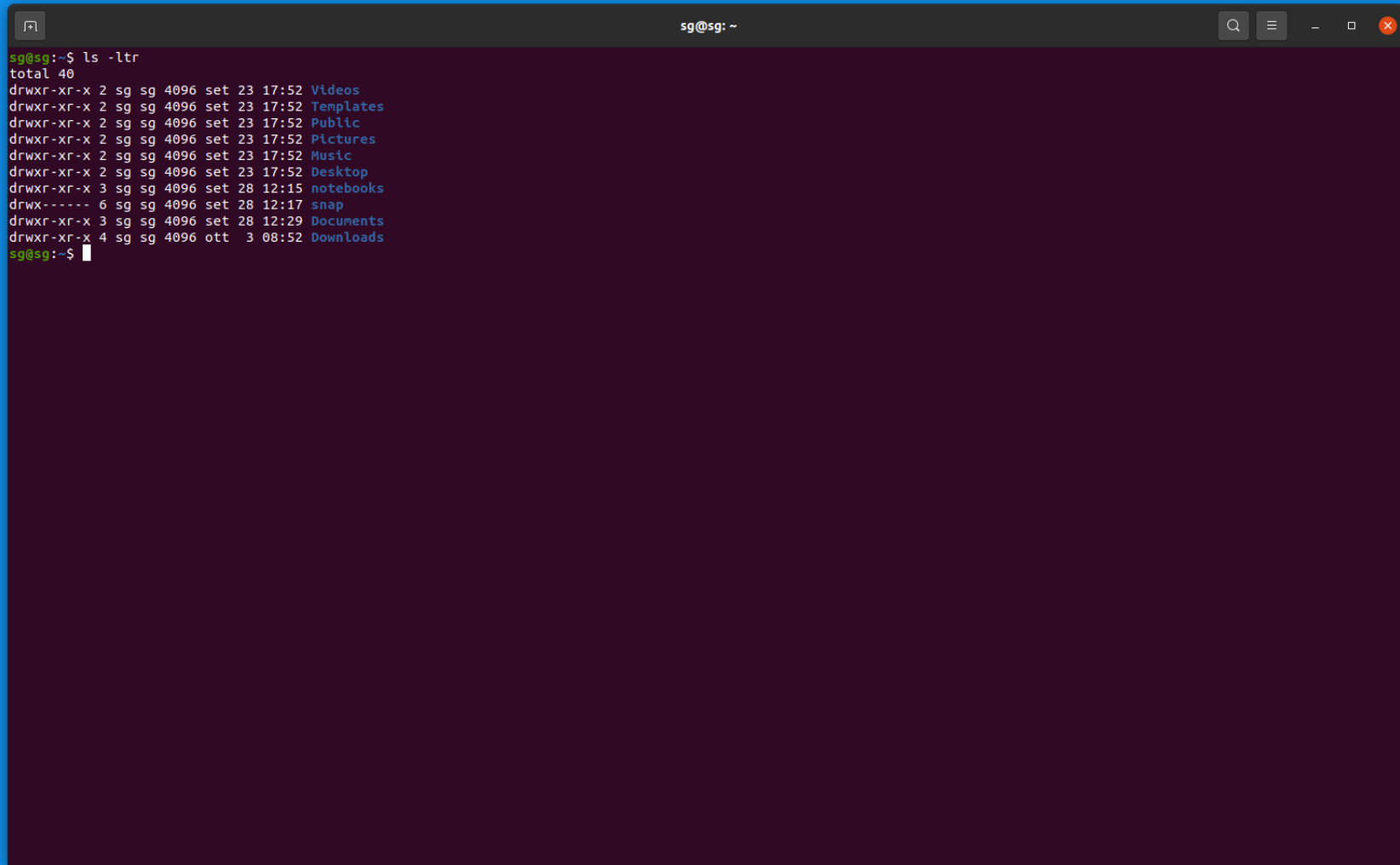
Linux



Linux

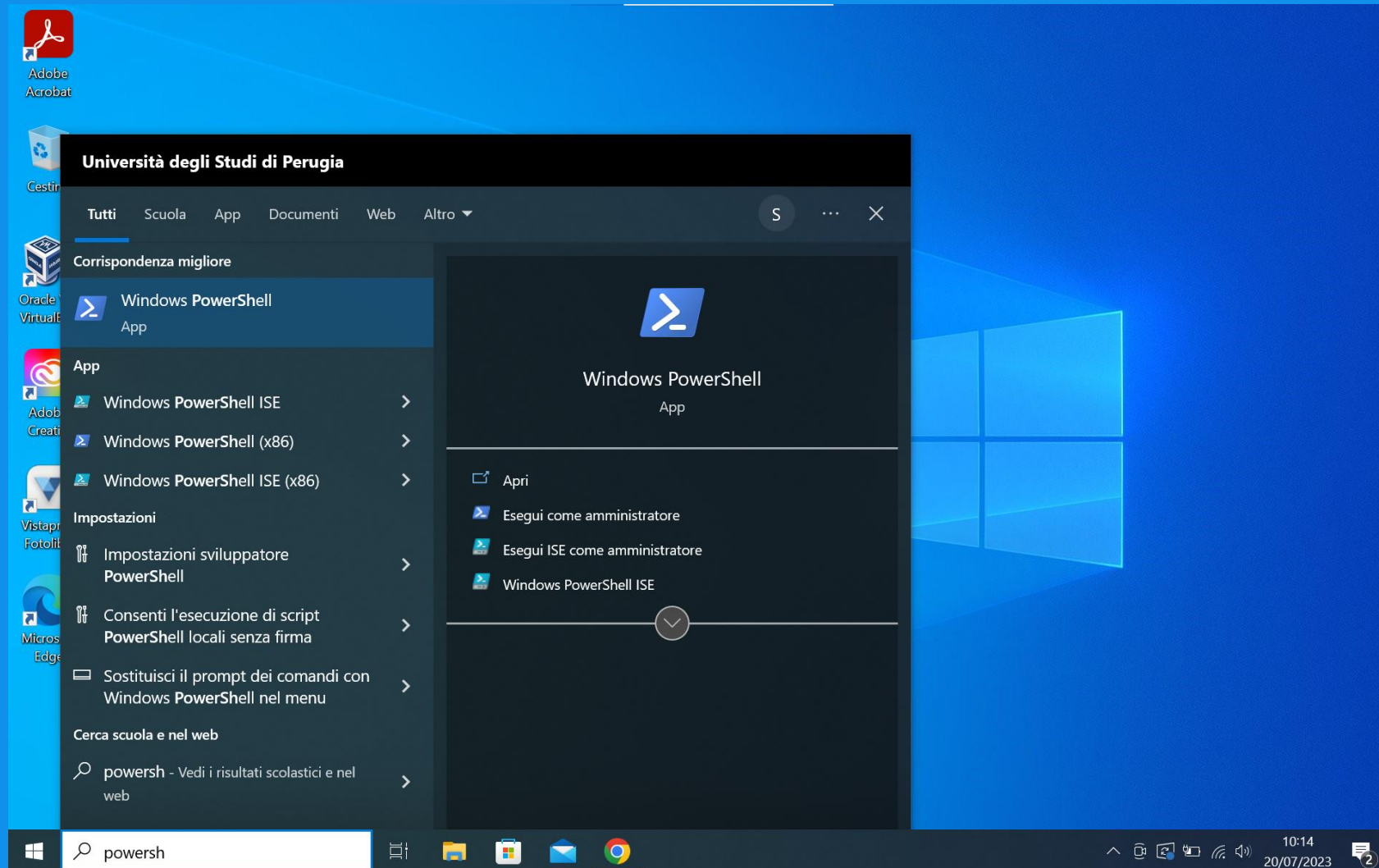


Terminale – Linux / Mac OS

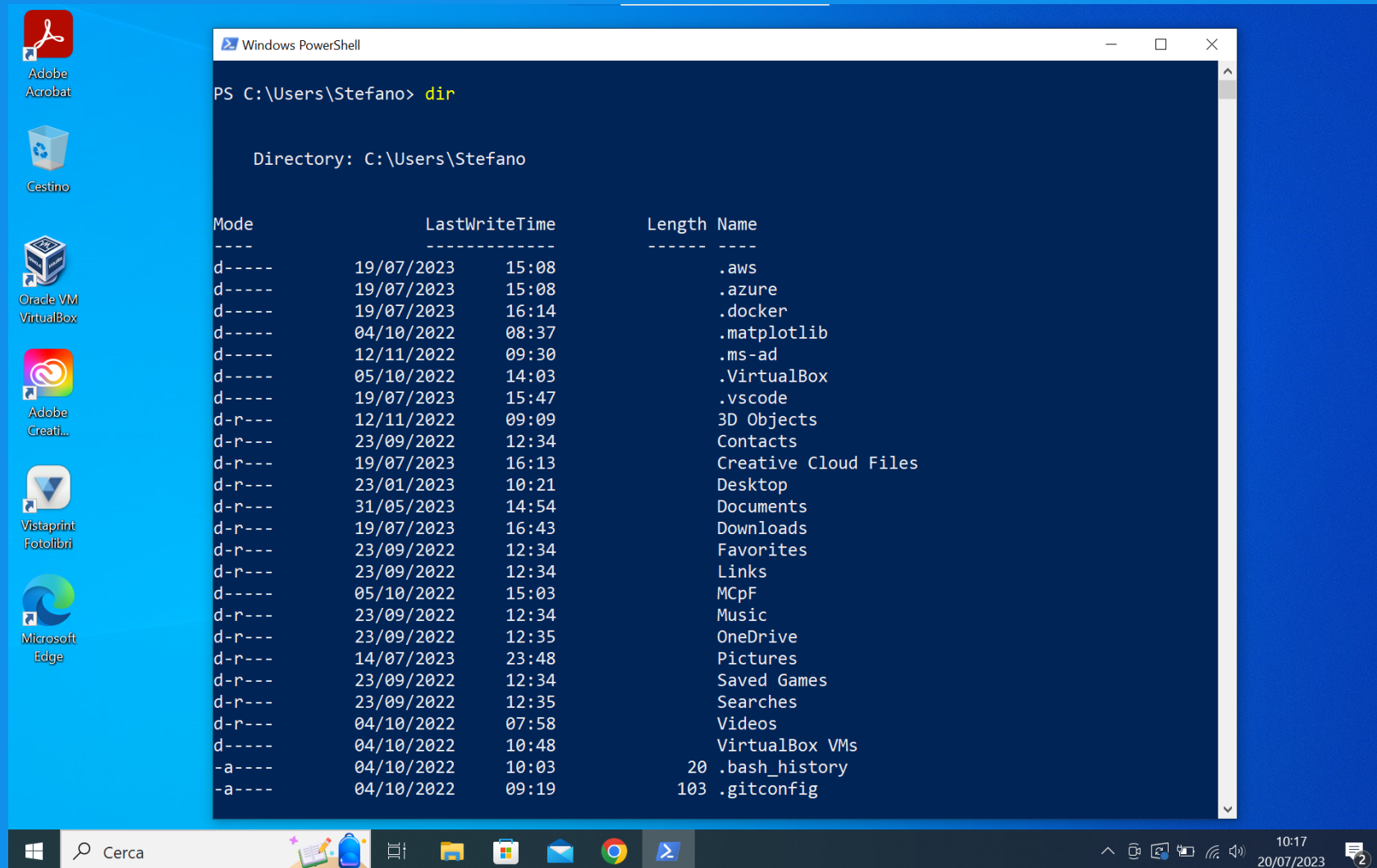


```
sg@sg:~$ ls -ltr
total 40
drwxr-xr-x 2 sg sg 4096 set 23 17:52 Videos
drwxr-xr-x 2 sg sg 4096 set 23 17:52 Templates
drwxr-xr-x 2 sg sg 4096 set 23 17:52 Public
drwxr-xr-x 2 sg sg 4096 set 23 17:52 Pictures
drwxr-xr-x 2 sg sg 4096 set 23 17:52 Music
drwxr-xr-x 2 sg sg 4096 set 23 17:52 Desktop
drwxr-xr-x 3 sg sg 4096 set 28 12:15 notebooks
drwx----- 6 sg sg 4096 set 28 12:17 snap
drwxr-xr-x 3 sg sg 4096 set 28 12:29 Documents
drwxr-xr-x 4 sg sg 4096 ott  3 08:52 Downloads
sg@sg:~$
```

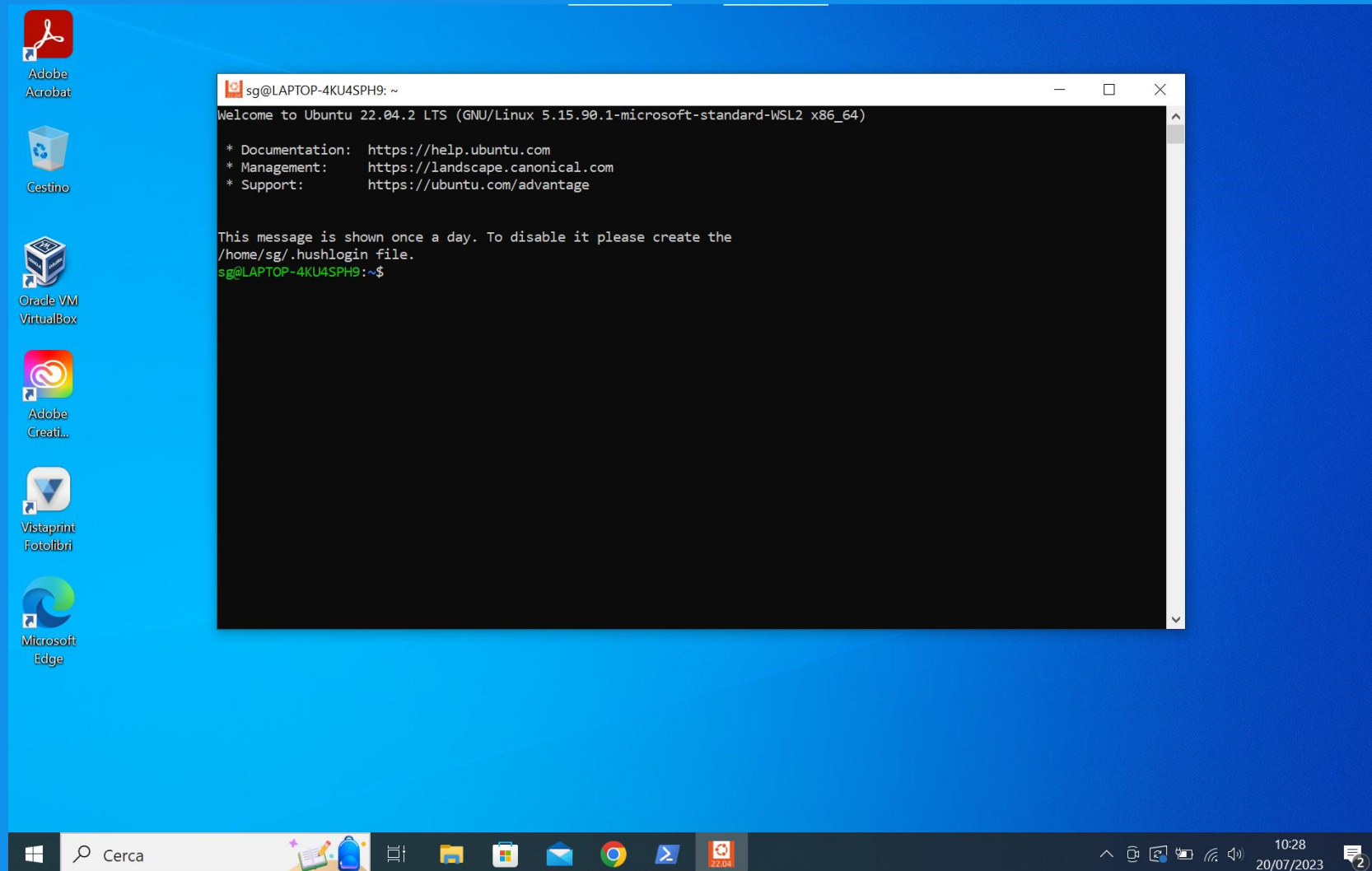
Windows PowerShell



Windows Power Shell



Windows Subsystem for Linux (WSL)



WSL

<https://learn.microsoft.com/it-it/windows/wsl/install>

The screenshot shows a web browser window displaying the Microsoft Learn page for installing Linux in Windows using WSL. The page is in Italian and features a navigation menu on the left, a main content area with the article title and introduction, and a right sidebar with additional resources.

Navigation Menu (Left):

- Windows
 - Integrità rilascio
 - Client Windows
 - Sviluppatori di applicazioni
 - Sviluppatori di hardware
 - Windows Server
 - Windows per IoT
 - Più informazioni
- Documentazione di WSL
 - Panoramica
 - Installazione
 - Installare WSL**
 - Passaggi di installazione manuali per le versioni precedenti
 - Installare in Windows Server
 - Esercitazioni
 - Concetti
 - Procedure
 - Domande frequenti
 - Risoluzione dei problemi
 - Note sulla versione
- Scarica il PDF

Main Content Area:

Learn / Windows / Ambiente di sviluppo / WSL /

Installare Linux in Windows con WSL

Articolo • 07/07/2023 • 8 contributori

[Commenti e suggerimenti](#)

In questo articolo

- [Prerequisiti](#)
- [Installare il comando WSL](#)
- [Modificare la distribuzione Linux predefinita installata](#)
- [Configurare le informazioni utente Linux](#)
- [Mostra 6 in più](#)

Gli sviluppatori possono accedere alla potenza di Windows e Linux contemporaneamente in un computer Windows. La sottosistema Windows per Linux (WSL) consente agli sviluppatori di installare una distribuzione Linux (ad esempio Ubuntu, OpenSUSE, Kali, Debian, Arch Linux e così via) e usare

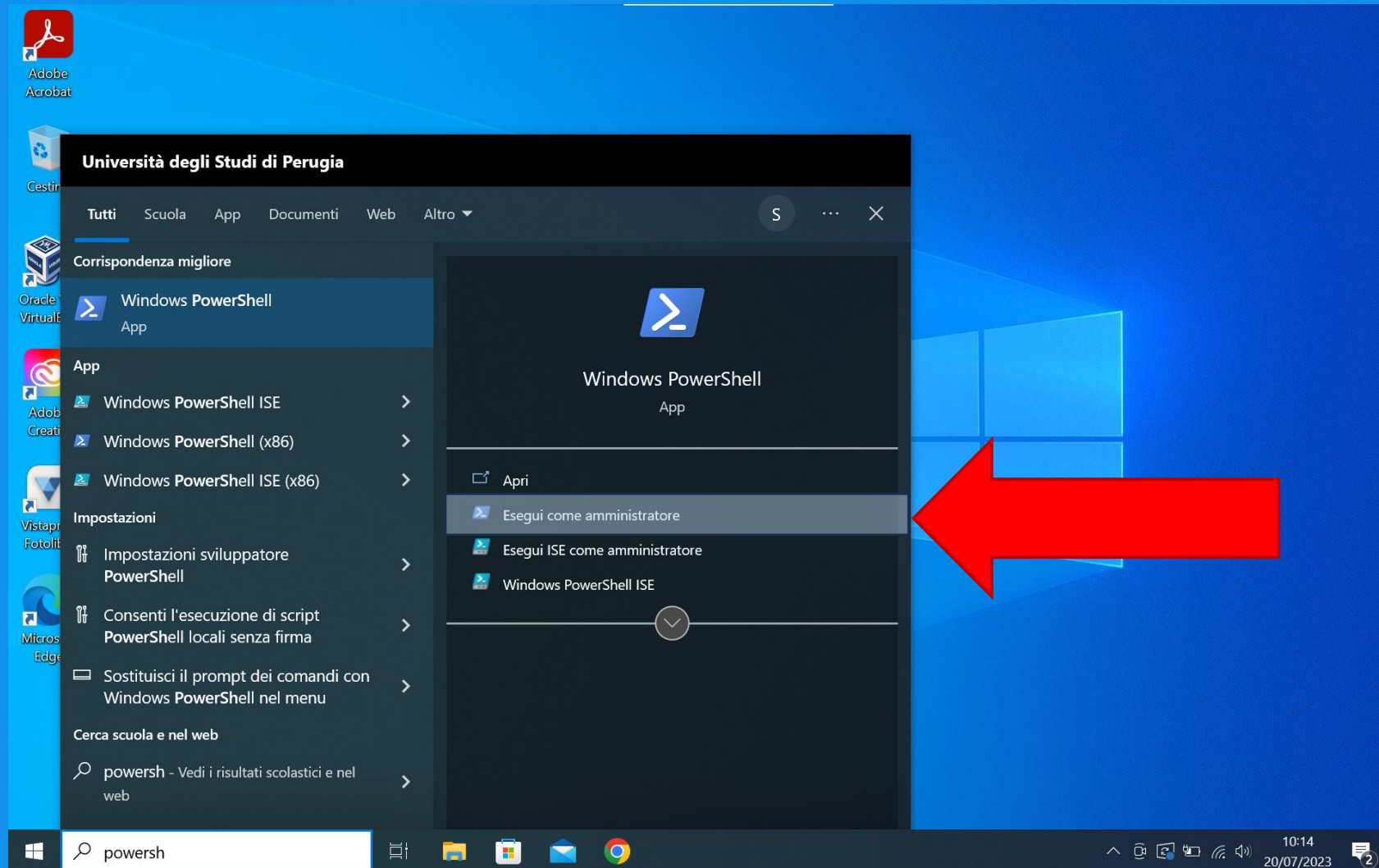
Right Sidebar:

Risorse aggiuntive

Documentazione

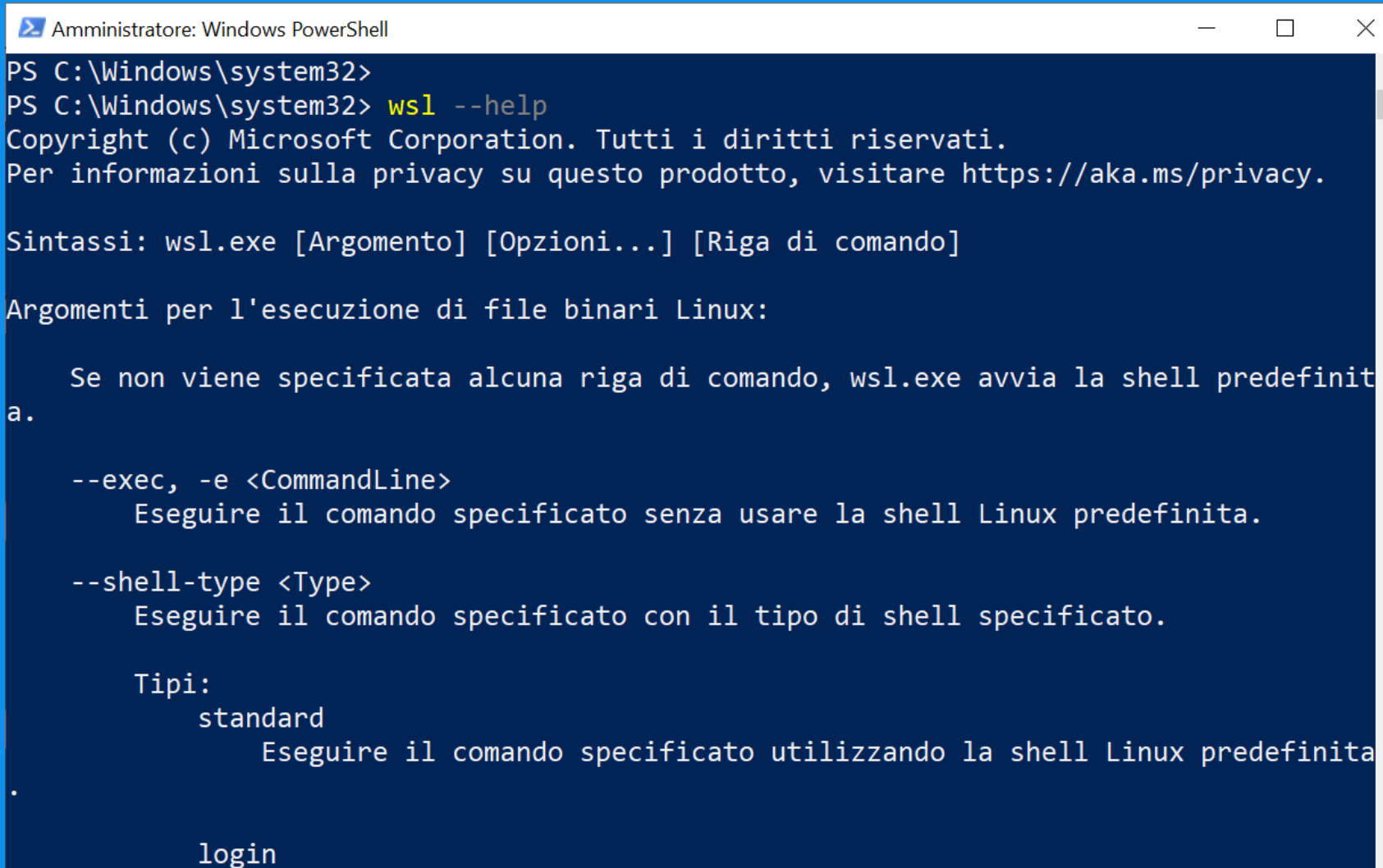
- [Configurare un ambiente di sviluppo WSL](#)
Configurare un ambiente di sviluppo WSL usando le procedure consigliate di questa guida dettagliata. Informazioni su come...
- [Confronto tra versioni di WSL](#)
WSL 2 offre i vantaggi di WSL 1, ma usa un kernel Linux effettivo, anziché un livello di conversione come WSL 1, con...
- [Risoluzione dei problemi relativi a Sottosistema Windows per Linux](#)
Fornisce informazioni dettagliate sugli errori e sui problemi comuni che si verificano durante l'esecuzione di Linux i...
- [Mostra 5 in più](#)

Windows PowerShell come Amministratore



Se richiedo concedere
il permesso di
apportare modifiche
al dispositivo

WSL

A screenshot of a Windows PowerShell terminal window titled "Amministratore: Windows PowerShell". The terminal shows the command `wsl --help` being executed. The output displays copyright information for Microsoft Corporation, a link to the privacy policy, and the syntax for the `wsl.exe` command. It also lists arguments for Linux binary execution, including `--exec` and `--shell-type`, and lists shell types: `standard` and `login`.

```
Amministratore: Windows PowerShell
PS C:\Windows\system32>
PS C:\Windows\system32> wsl --help
Copyright (c) Microsoft Corporation. Tutti i diritti riservati.
Per informazioni sulla privacy su questo prodotto, visitare https://aka.ms/privacy.

Sintassi: wsl.exe [Argomento] [Opzioni...] [Riga di comando]

Argomenti per l'esecuzione di file binari Linux:

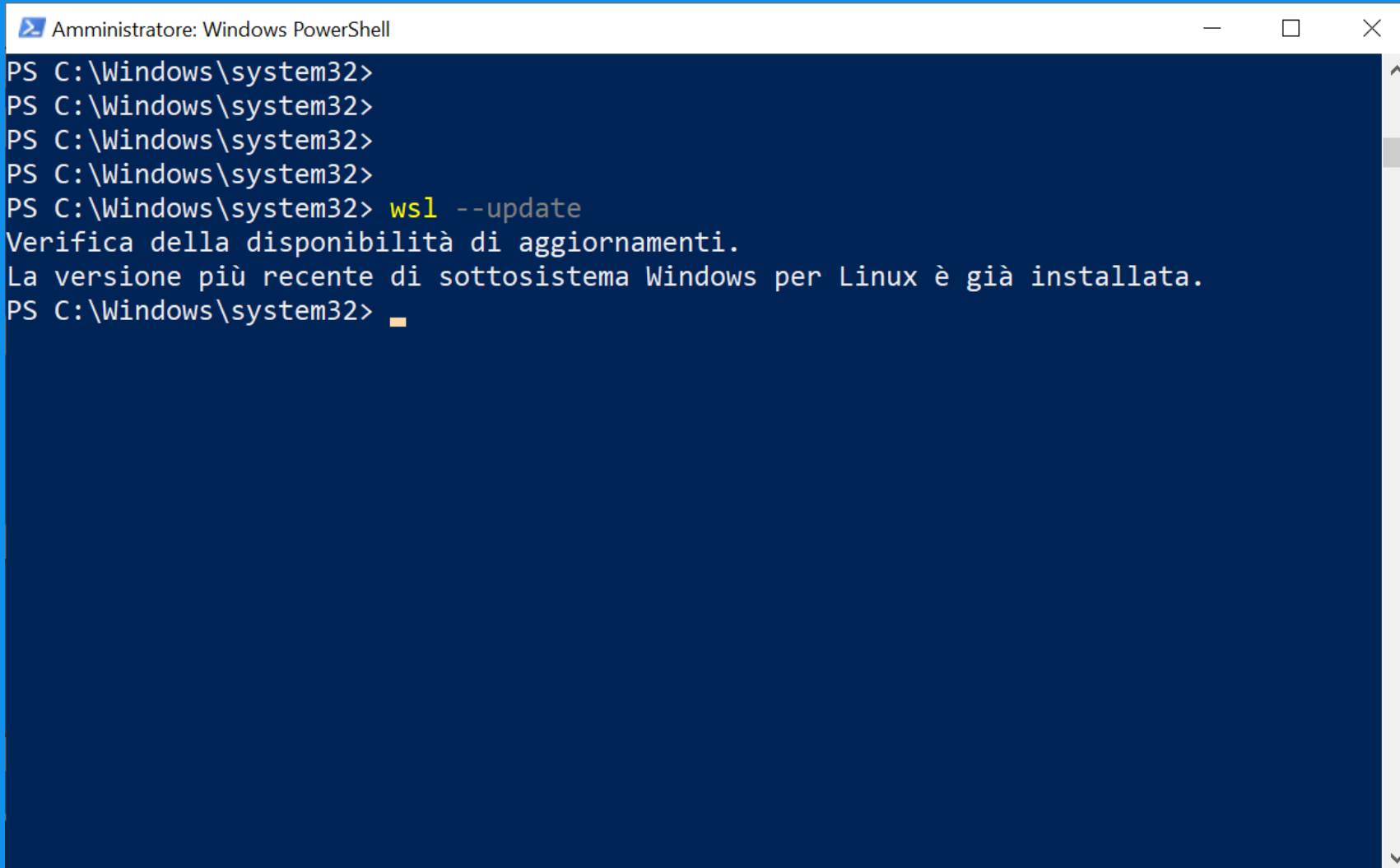
    Se non viene specificata alcuna riga di comando, wsl.exe avvia la shell predefinita.

    --exec, -e <CommandLine>
        Eseguire il comando specificato senza usare la shell Linux predefinita.

    --shell-type <Type>
        Eseguire il comando specificato con il tipo di shell specificato.

    Tipi:
        standard
            Eseguire il comando specificato utilizzando la shell Linux predefinita.
        login
```

WSL



```
Amministratore: Windows PowerShell
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> wsl --update
Verifica della disponibilità di aggiornamenti.
La versione più recente di sottosistema Windows per Linux è già installata.
PS C:\Windows\system32> █
```

WSL

Aprire PowerShell come Amministratore

```
Amministratore: Windows PowerShell
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> wsl -l -o
Di seguito è riportato un elenco di distribuzioni valide che è possibile installare.
Installare con 'wsl.exe --install <Distro>'.

NAME                                FRIENDLY NAME
Ubuntu                              Ubuntu
Ubuntu-26.04                        Ubuntu 26.04 LTS
Ubuntu-24.04                        Ubuntu 24.04 LTS
Ubuntu-22.04                        Ubuntu 22.04 LTS
openSUSE-Tumbleweed                 openSUSE Tumbleweed
openSUSE-Leap-16.0                  openSUSE Leap 16.0
SUSE-Linux-Enterprise-15-SP7        SUSE Linux Enterprise 15 SP7
SUSE-Linux-Enterprise-16.0          SUSE Linux Enterprise 16.0
kali-linux                          Kali Linux Rolling
Debian                              Debian GNU/Linux
AlmaLinux-8                         AlmaLinux OS 8
AlmaLinux-9                         AlmaLinux OS 9
AlmaLinux-Kitten-10                 AlmaLinux OS Kitten 10
AlmaLinux-10                       AlmaLinux OS 10
archlinux                           Arch Linux
FedoraLinux-44                      Fedora Linux 44
FedoraLinux-43                      Fedora Linux 43
eLxr                                eLxr 12.12.0.0 GNU/Linux
OracleLinux_7_9                     Oracle Linux 7.9
OracleLinux_8_10                    Oracle Linux 8.10
OracleLinux_9_5                     Oracle Linux 9.5
SUSE-Linux-Enterprise-15-SP6        SUSE Linux Enterprise 15 SP6
PS C:\WINDOWS\system32>
```

WSL

Aprire PowerShell come Amministratore

```
Amministratore: Windows PowerShell
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> wsl -l -o
Di seguito è riportato un elenco di distribuzioni valide che è possibile installare.
Installare con 'wsl.exe --install <Distro>'.

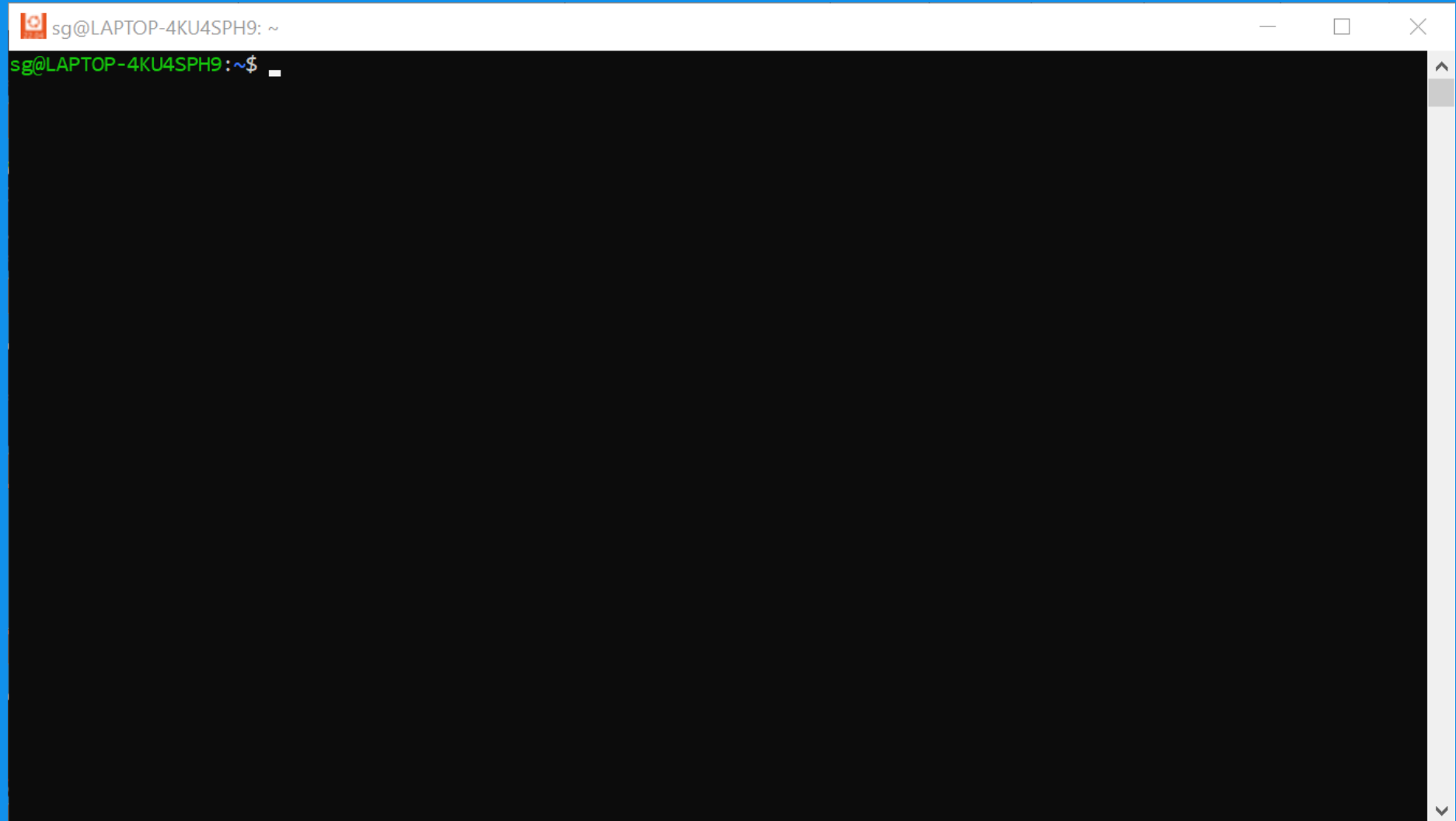
NAME                                FRIENDLY NAME
-----
Ubuntu                              Ubuntu
Ubuntu-26.04                        Ubuntu 26.04 LTS
Ubuntu-24.04                        Ubuntu 24.04 LTS
Ubuntu-22.04                        Ubuntu 22.04 LTS
openSUSE-Tumbleweed                 openSUSE Tumbleweed
openSUSE-Leap-16.0                  openSUSE Leap 16.0
SUSE-Linux-Enterprise-15-SP7        SUSE Linux Enterprise 15 SP7
SUSE-Linux-Enterprise-16.0          SUSE Linux Enterprise 16.0
kali-linux                          Kali Linux Rolling
Debian                              Debian GNU/Linux
AlmaLinux-8                         AlmaLinux OS 8
AlmaLinux-9                         AlmaLinux OS 9
AlmaLinux-Kitten-10                 AlmaLinux OS Kitten 10
AlmaLinux-10                        AlmaLinux OS 10
archlinux                           Arch Linux
FedoraLinux-44                      Fedora Linux 44
FedoraLinux-43                      Fedora Linux 43
eLxr                                eLxr 12.12.0.0 GNU/Linux
OracleLinux_7_9                     Oracle Linux 7.9
OracleLinux_8_10                    Oracle Linux 8.10
OracleLinux_9_5                     Oracle Linux 9.5
SUSE-Linux-Enterprise-15-SP6        SUSE Linux Enterprise 15 SP6
PS C:\WINDOWS\system32>
```

WSL

```
Amministratore: Windows PowerShell
PS C:\WINDOWS\system32> wsl -l -o
Di seguito è riportato un elenco di distribuzioni valide che è possibile installare.
Installare con 'wsl.exe --install <Distro>'.

NAME                                FRIENDLY NAME
Ubuntu                             Ubuntu
Ubuntu-26.04                       Ubuntu 26.04 LTS
Ubuntu-24.04                       Ubuntu 24.04 LTS
Ubuntu-22.04                       Ubuntu 22.04 LTS
openSUSE-Tumbleweed                openSUSE Tumbleweed
openSUSE-Leap-16.0                 openSUSE Leap 16.0
SUSE-Linux-Enterprise-15-SP7       SUSE Linux Enterprise 15 SP7
SUSE-Linux-Enterprise-16.0         SUSE Linux Enterprise 16.0
kali-linux                        Kali Linux Rolling
Debian                             Debian GNU/Linux
AlmaLinux-8                       AlmaLinux OS 8
AlmaLinux-9                       AlmaLinux OS 9
AlmaLinux-Kitten-10               AlmaLinux OS Kitten 10
AlmaLinux-10                      AlmaLinux OS 10
archlinux                         Arch Linux
FedoraLinux-44                    Fedora Linux 44
FedoraLinux-43                    Fedora Linux 43
eLxR                              eLxR 12.12.0.0 GNU/Linux
OracleLinux_7_9                   Oracle Linux 7.9
OracleLinux_8_10                  Oracle Linux 8.10
OracleLinux_9_5                   Oracle Linux 9.5
SUSE-Linux-Enterprise-15-SP6       SUSE Linux Enterprise 15 SP6
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> wsl --install -d Ubuntu-24.06
```

Terminale Linux WLS



Comandi Base Terminale Linux - pwd



The screenshot shows a terminal window with the title bar "sg@LAPTOP-4KU4SPH9: ~". The terminal content consists of 20 lines of the command "pwd" followed by the output "/home/sg". The text "Posizione attuale nell'albero delle cartelle" is overlaid on the right side of the terminal output.

```
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
sg@LAPTOP-4KU4SPH9: ~$ pwd
/home/sg
```

Posizione attuale nell'albero delle cartelle

Comandi Base Terminale Linux – cd ..

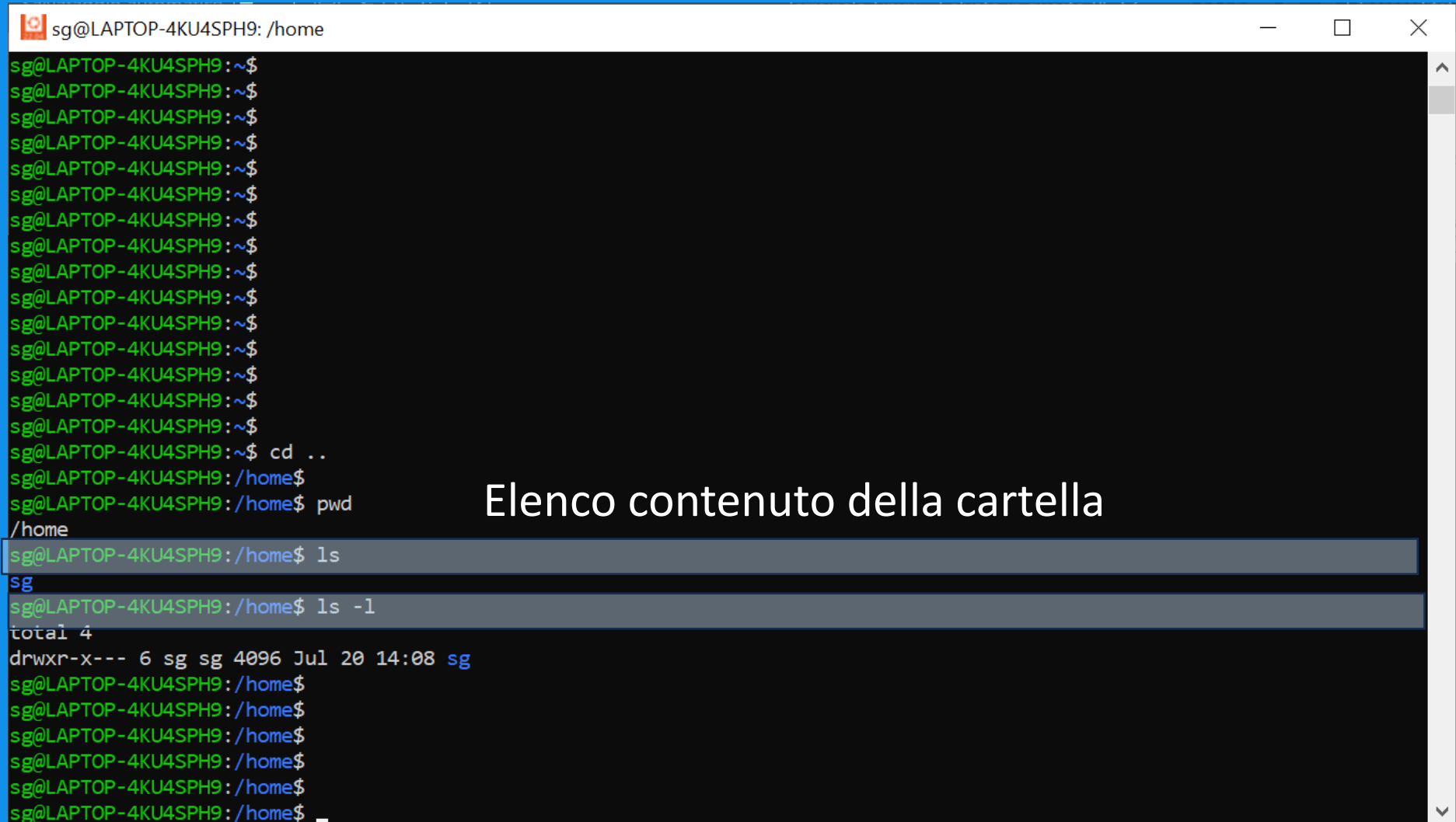
[illegible]

Spostamento nella cartella superiore

• • •

```
sg@LAPTOP-4KU4SPH9: /home
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$ cd ..
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$ pwd
/home
sg@LAPTOP-4KU4SPH9: /home$
```

Comandi Base Terminale Linux – ls



```
sg@LAPTOP-4KU4SPH9: /home
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$ cd ..
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$ pwd
/home
sg@LAPTOP-4KU4SPH9: /home$ ls
sg
sg@LAPTOP-4KU4SPH9: /home$ ls -l
total 4
drwxr-x--- 6 sg sg 4096 Jul 20 14:08 sg
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$
sg@LAPTOP-4KU4SPH9: /home$
```

Elenco contenuto della cartella

... Help

```
sg@LAPTOP-4KU4SPH9: /home
sg@LAPTOP-4KU4SPH9:/home$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILES (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                do not ignore entries starting with .
  -A, --almost-all        do not list implied . and ..
      --author              with -l, print the author of each file
  -b, --escape             print C-style escapes for nongraphic characters
      --block-size=SIZE    with -l, scale sizes by SIZE when printing them;
                           e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups     do not list implied entries ending with ~
  -c                       with -lt: sort by, and show, ctime (time of last
                           modification of file status information);
                           with -l: show ctime and sort by name;
                           otherwise: sort by ctime, newest first
  -C                       list entries by columns
      --color[=WHEN]       colorize the output; WHEN can be 'always' (default
                           if omitted), 'auto', or 'never'; more info below
  -d, --directory          list directories themselves, not their contents
  -D, --dired              generate output designed for Emacs' dired mode
  -f                       do not sort, enable -aU, disable -ls --color
  -F, --classify           append indicator (one of */=>@|) to entries
                           likewise, except do not append '*'
      --format=WORD         across -x, commas -m, horizontal -x, long -l,
                           single-column -1, verbose -l, vertical -C
      --full-time           like -l --time-style=full-iso
  -g                       like -l, but do not list owner
      --group-directories-first
```

Comandi Base Terminale Linux - *cd dir*

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ cd ..  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$ pwd  
/home  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$ ls -l  
total 4  
drwxr-x--- 7 sg sg 4096 Jul 26 18:38 sg  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$ cd sg/  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ pwd  
/home/sg  
sg@LAPTOP-4KU4SPH9:~$
```

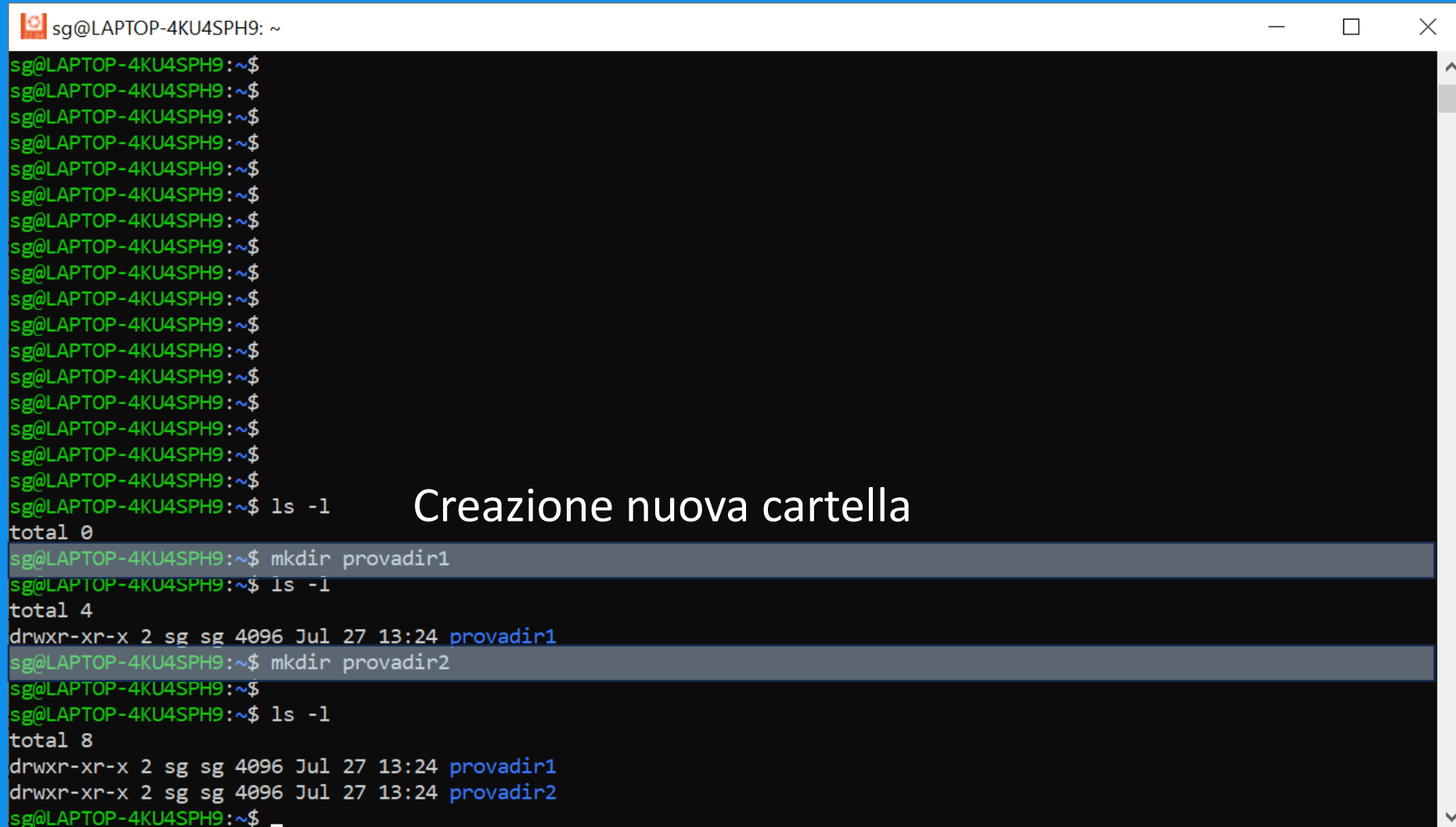
Spostamento in una cartella specifica

Comandi Base Terminale Linux – cd -

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ cd ..  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$ pwd  
/home  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$ ls -l  
total 4  
drwxr-x--- 7 sg sg 4096 Jul 26 18:38 sg  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$  
sg@LAPTOP-4KU4SPH9:/home$ cd sg/  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ pwd  
/home/sg  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ cd -  
/home  
sg@LAPTOP-4KU4SPH9:/home$ pwd  
/home  
sg@LAPTOP-4KU4SPH9:/home$ cd -  
/home/sg  
sg@LAPTOP-4KU4SPH9:~$ pwd  
/home/sg  
sg@LAPTOP-4KU4SPH9:~$
```

Spostamento nella cartella precedente

Comandi Base Terminale Linux – mkdir



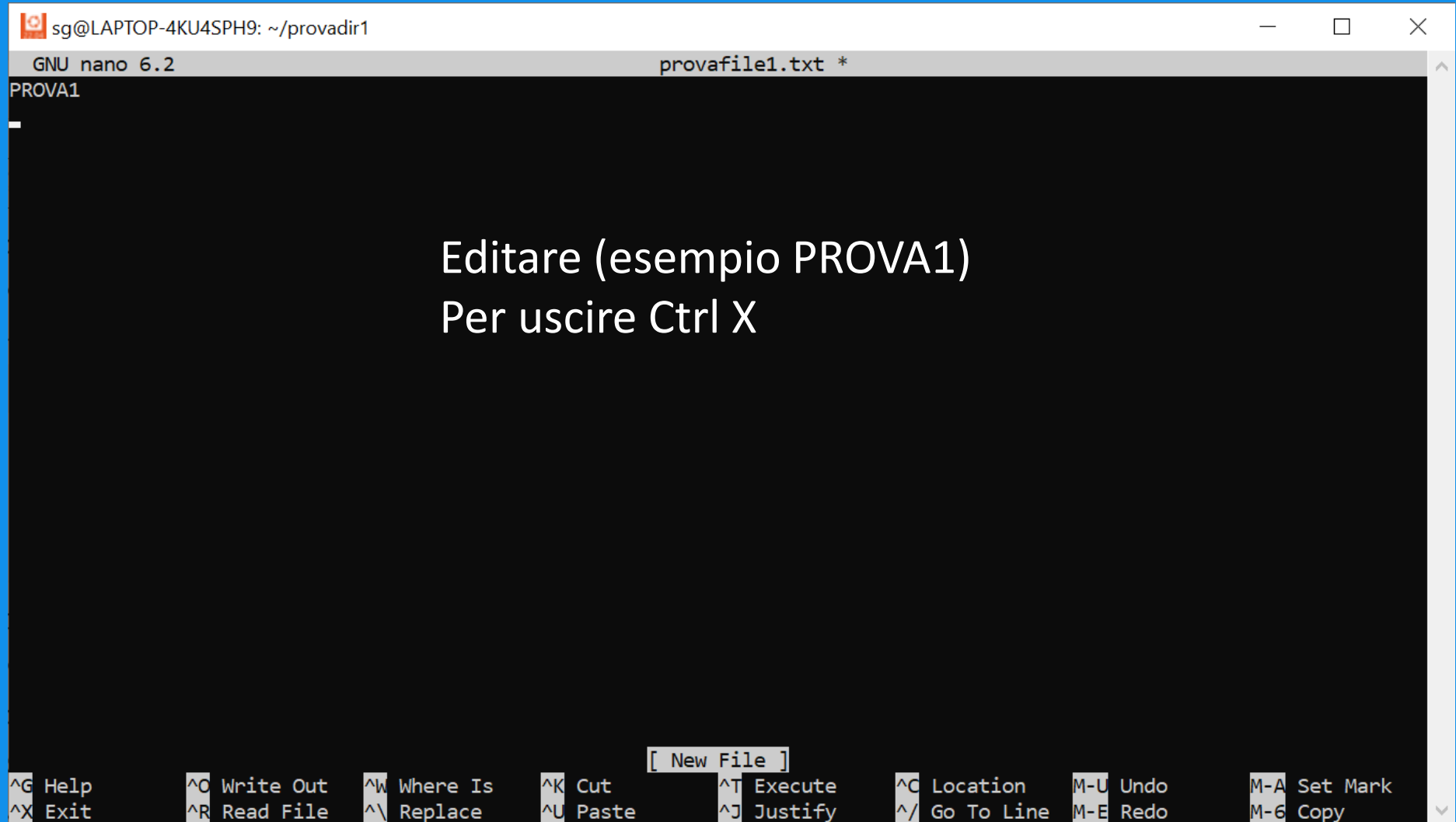
```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls -l  
total 0  
sg@LAPTOP-4KU4SPH9:~$ mkdir provadir1  
sg@LAPTOP-4KU4SPH9:~$ ls -l  
total 4  
drwxr-xr-x 2 sg sg 4096 Jul 27 13:24 provadir1  
sg@LAPTOP-4KU4SPH9:~$ mkdir provadir2  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls -l  
total 8  
drwxr-xr-x 2 sg sg 4096 Jul 27 13:24 provadir1  
drwxr-xr-x 2 sg sg 4096 Jul 27 13:24 provadir2  
sg@LAPTOP-4KU4SPH9:~$
```

Creazione nuova cartella

Comandi Base Terminale Linux – pico

[illegible]

Comandi Base Terminale Linux – pico



```
sg@LAPTOP-4KU4SPH9: ~/provalidir1
GNU nano 6.2 provafile1.txt *
PROVA1
-

Editare (esempio PROVA1)
Per uscire Ctrl X

[ New File ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-6 Copy
```

```
sg@LAPTOP-4KU4SPH9: ~/provadir1  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$ pico provafire1.txt  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$  
sg@LAPTOP-4KU4SPH9:~/provadir1$ ls -l  
total 4  
-rw-r--r-- 1 sg sg 8 Jul 27 20:44 provafire1.txt  
sg@LAPTOP-4KU4SPH9:~/provadir1$
```

Comandi Base Terminale Linux - more

[illegible]

Comandi Base Terminale Linux - rm

```
sg@LAPTOP-4KU4SPH9: ~/provadir1
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$ ls -l
total 4
-rw-r--r-- 1 sg sg 8 Jul 27 20:44 provafile1.txt
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$ rm provafile1.txt
sg@LAPTOP-4KU4SPH9:~/provadir1$
sg@LAPTOP-4KU4SPH9:~/provadir1$ ls -l
total 0
sg@LAPTOP-4KU4SPH9:~/provadir1$
```

Rimozione file

Comandi Base Terminale Linux - rm

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ pwd  
/home/sg  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls -l  
total 8  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
drwxr-xr-x 2 sg sg 4096 Jul 27 21:01 provadir2  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ rm provadir2  
rm: cannot remove 'provadir2': Is a directory  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ rm -R provadir2  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls -l  
total 4  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
sg@LAPTOP-4KU4SPH9:~$
```

Rimozione cartella

Accesso Incrociato WSL-Windows

[illegible]

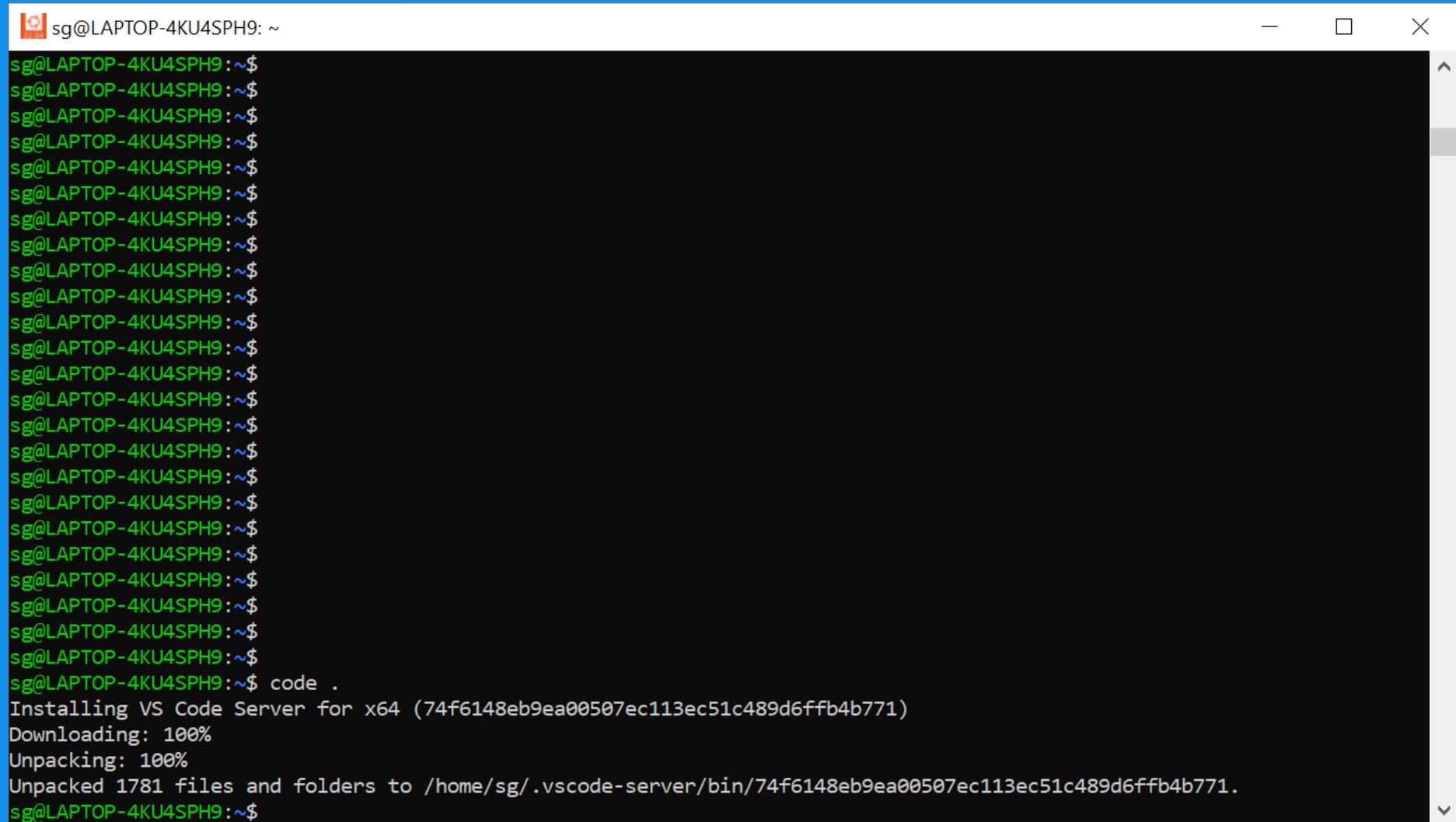
Accesso Incrociato WSL-Windows

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls /mnt/c/  
ls: cannot access '/mnt/c/DumpStack.log.tmp': Permission denied  
ls: cannot access '/mnt/c/hiberfil.sys': Permission denied  
ls: cannot access '/mnt/c/pagefile.sys': Permission denied  
ls: cannot access '/mnt/c/swapfile.sys': Permission denied  
'$Recycle.Bin'          DumpStack.log          'Program Files'        'System Volume Information'  pagefile.sys  
'$WinREAgent'          DumpStack.log.tmp      'Program Files (x86)'  Users                        swapfile.sys  
DRIVERS                Intel                  ProgramData             Windows  
'Documents and Settings' PerfLogs              Recovery               hiberfil.sys  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ explorer.exe .  
sg@LAPTOP-4KU4SPH9:~$
```

...

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls /mnt/c/  
ls: cannot access '/mnt/c/DumpStack.log.tmp': Permission denied  
ls: cannot access '/mnt/c/hiberfil.sys': Permission denied  
ls: cannot access '/mnt/c/pagefile.sys': Permission denied  
ls: cannot access '/mnt/c/swapfile.sys': Permission denied  
'$Recycle.Bin'      DumpStack.log      'Program Files'      'System Volume Information'  pagefile.sys  
'$WinREAgent'      DumpStack.log.tmp  'Program Files (x86)'  Users                      swapfile.sys  
DRIVERS            Intel              ProgramData           Windows  
'Documents and Settings' PerfLogs          Recovery              hiberfil.sys  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls /mnt/c/Users/  
'All Users'  Default  'Default User'  Public  Stefano  desktop.ini  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$
```

WSL – Visual Studio Code (da Windows)



The screenshot shows a terminal window titled "sg@LAPTOP-4KU4SPH9: ~". The terminal displays a series of 20 identical prompts "sg@LAPTOP-4KU4SPH9: ~\$". After the last prompt, the user enters the command "code .". The terminal then shows the following output:

```
Installing VS Code Server for x64 (74f6148eb9ea00507ec113ec51c489d6ffb4b771)
Downloading: 100%
Unpacking: 100%
Unpacked 1781 files and folders to /home/sg/.vscode-server/bin/74f6148eb9ea00507ec113ec51c489d6ffb4b771.
sg@LAPTOP-4KU4SPH9: ~$
```

Ubuntu 24.04

```
#echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf > /dev/null (NON necessario)
```

```
sudo apt update
```

```
sudo apt install emacs git
```

```
sudo apt-get install wget
```

```
sudo apt install python3-dev python3-tk python3-pyqt5
```

```
sudo apt install python3-pip
```

```
sudo apt install python3-numpy python3-scipy python3-matplotlib python3-pandas  
python3-sympy
```

```
python3 -m pip install --break-system-packages argparse
```

Mac OS

```
python3 -m ensurepip
```

In alternativa bisogna usare il manager di pacchetti Homebrew (<https://brew.sh/>):

```
brew install python
```

```
python3 -m pip install numpy scipy matplotlib pandas sympy
```

```
python3 -m pip install --break-system-packages argparse
```

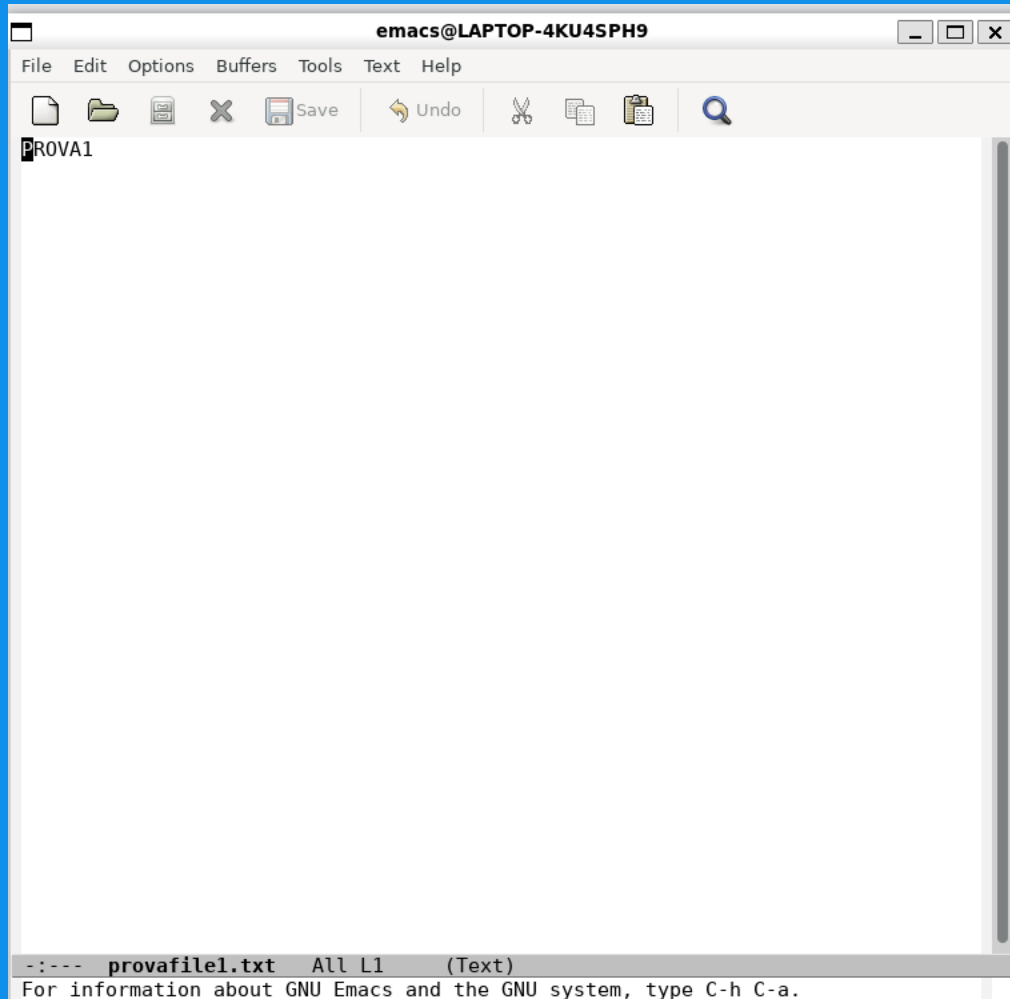
Homebrew (<https://brew.sh/>) è utile anche per installare altri pacchetti:

- `brew install wget`
- `brew install git`

Comandi Base Terminale Linux – emacs

[illegible]

Comandi Base Terminale Linux – emacs



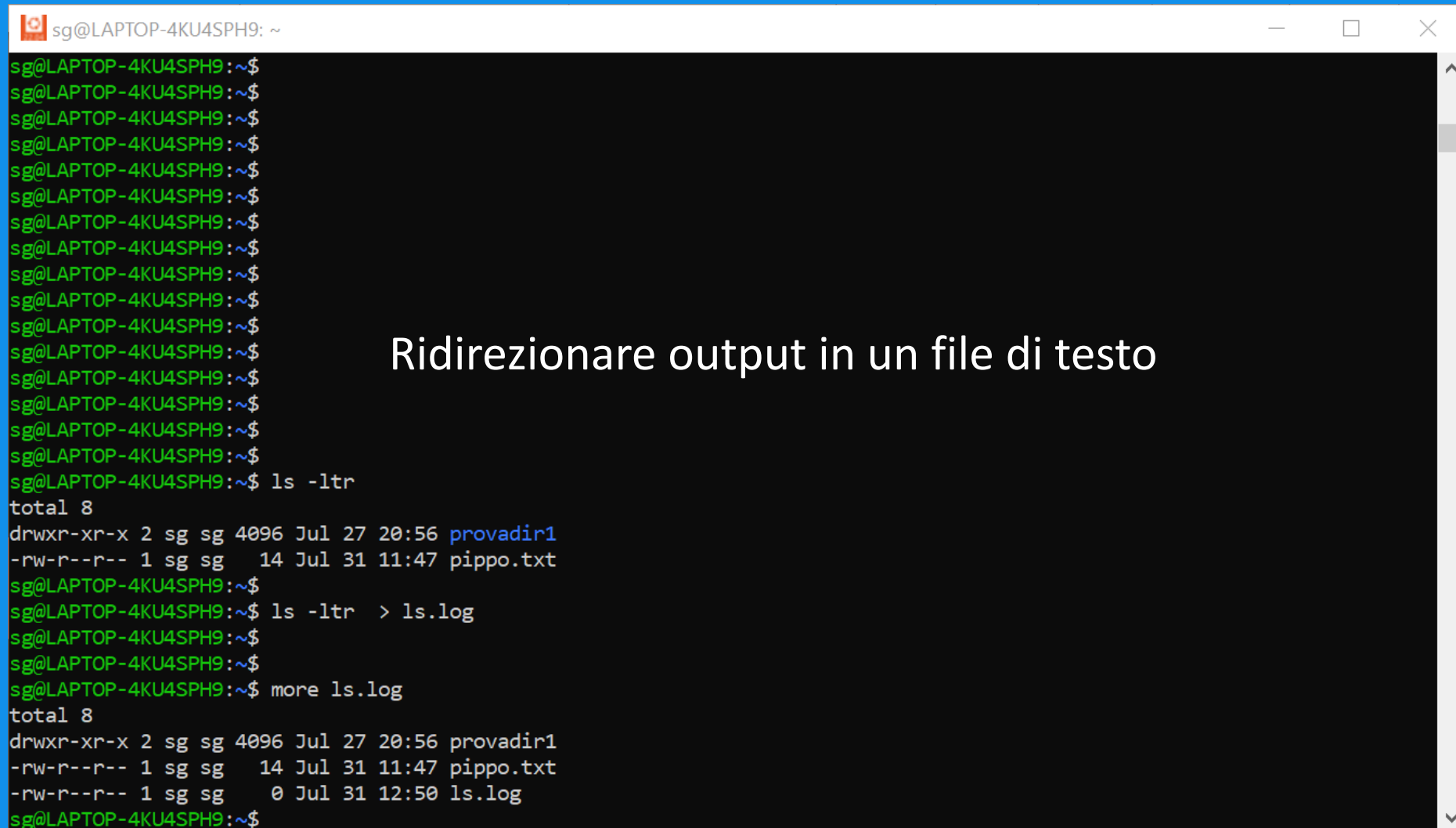
Editare

Per salvare: Ctrl x Ctrl s

Per uscire: Ctrl x Ctrl c

+ molti altri comandi

Comandi Base Terminale Linux – >, >>



The screenshot shows a terminal window titled "sg@LAPTOP-4KU4SPH9: ~". The user enters the command `ls -ltr`, which lists files in the current directory. The output shows two files: `provadir1` and `pippo.txt`. Then, the user enters `ls -ltr > ls.log`, which redirects the output of the `ls -ltr` command to a file named `ls.log`. Finally, the user enters `more ls.log`, which displays the contents of `ls.log`, showing the same output as the previous `ls -ltr` command, but now including the `ls.log` file itself.

```
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ ls -ltr  
total 8  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
-rw-r--r-- 1 sg sg  14 Jul 31 11:47 pippo.txt  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ ls -ltr > ls.log  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$  
sg@LAPTOP-4KU4SPH9: ~$ more ls.log  
total 8  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
-rw-r--r-- 1 sg sg  14 Jul 31 11:47 pippo.txt  
-rw-r--r-- 1 sg sg   0 Jul 31 12:50 ls.log  
sg@LAPTOP-4KU4SPH9: ~$
```

Ridirezionare output in un file di testo

Comandi Base Terminale Linux – >, >>

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ more ls.log  
total 8  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
-rw-r--r-- 1 sg sg  14 Jul 31 11:47 pippo.txt  
-rw-r--r-- 1 sg sg   0 Jul 31 12:50 ls.log  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls -ltr >> ls.log  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ more ls.log  
total 8  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
-rw-r--r-- 1 sg sg  14 Jul 31 11:47 pippo.txt  
-rw-r--r-- 1 sg sg   0 Jul 31 12:50 ls.log  
total 12  
drwxr-xr-x 2 sg sg 4096 Jul 27 20:56 provadir1  
-rw-r--r-- 1 sg sg  14 Jul 31 11:47 pippo.txt  
-rw-r--r-- 1 sg sg 146 Jul 31 12:50 ls.log  
sg@LAPTOP-4KU4SPH9:~$
```

Appendere output ad un file di testo

Esercizio

1. Aprire il terminale Linux (Nativo o WSL);
2. Assicurarsi di essere nella cartella *home* dell'utente;
3. Creare la cartella *MCF* e spostarsi al suo interno;
4. Creare il file di testo *Nome_Cognome.txt* (sostituendo opportunamente);
5. Editare il file inserendo le seguenti informazioni:
 1. Nome;
 2. Cognome;
 3. Matricola;
 4. e-mail istituzionale;
6. Appendere output del comando *python3 --version* al file;
7. Inviare il file al docente via e-mail

ROOT

<https://root.cern/>

The screenshot shows the ROOT Data Analysis Framework website. The browser's address bar displays 'root.cern'. The website's header includes the ROOT logo and a navigation menu with links to 'About', 'Learn', 'Install', 'Forum & Help', 'Manual', 'Blog Posts', 'Contribute', and 'For Developers'. The main content area features a large banner with the text 'ROOT: analyzing petabytes of data, scientifically.' and a subtext 'An open-source data analysis framework used by high energy physics and others.' Below the banner are two buttons: 'Learn more' and 'Install v6.32.04'. A row of four icons with labels is positioned below the buttons: 'Learn' (a book icon), 'Reference' (a document icon), 'Forum' (a speech bubble icon), and 'Gallery' (a bar chart icon). At the bottom of the page, there are three decorative icons: a square root symbol with a minus sign and a 1, a globe, and a dollar sign followed by an underscore.

ROOT: analyzing petabytes of data, scientifically.

An open-source data analysis framework used by high energy physics and others.

[Learn more](#) [Install v6.32.04](#)

[Learn](#) [Reference](#) [Forum](#) [Gallery](#)

$\sqrt{-1}$  \$ _

ROOT

<https://root.cern/>

The screenshot shows the ROOT Data Analysis Framework website. The browser's address bar displays 'root.cern'. The website's header includes the ROOT logo and navigation links: 'About', 'Install', 'Manual', 'Contribute', 'For Developers', and 'Source'. The main heading reads 'ROOT: analyzing petabytes of data, scientifically.', followed by the subtitle 'An open-source data analysis framework used by high energy physics and others.' Below this, there are two buttons: 'Learn more' and 'Install v6.36.04'. A large background image depicts a particle detector structure with orange lines radiating from a central point. Below the main content, there are four icons with labels: a graduation cap for 'Learn', an open book for 'Reference', a speech bubble for 'Forum', and a scatter plot for 'Gallery'. At the bottom of the page, there are three more icons: a bar chart, a code block, and a dollar sign followed by an underscore.

ROOT: analyzing petabytes of data, scientifically.

An open-source data analysis framework used by high energy physics and others.

[Learn more](#) [Install v6.36.04](#)

 Learn

 Reference

 Forum

 Gallery

ROOT

<https://root.cern/>

Installing ROOT

ROOT is available on Linux, Mac, and Windows. To install it, we highly recommend (ordered by priority) either:

1. Use a package manager
2. Download a pre-compiled binary

Further ways to install ROOT on your computer are listed in the table of content on the right. Which one is best for you depends on your operating system and usage requirements.

In all cases, make sure to always use the most recent ROOT release possible to get the latest bug fixes, features and quick user support. The latest stable ROOT release is 6.32.04 (about ROOT versioning scheme).

Download a pre-compiled binary distribution

We distribute pre-compiled ROOT for several major Linux distributions as well as MacOS and (as a beta) Windows. The steps to install a pre-compiled binary are simple:

1. Install all [required dependencies](#) with the system package manager
2. [Download the release](#) for the desired platform and ROOT version
3. Unpack the archive
4. Add the ROOT libraries and executables to your environment by sourcing the appropriate `thisroot.*` script. These setup scripts can be found in the ROOT binary release, in the `bin` directory.

ROOT - Prerequisiti

```
sudo apt-get install libtbb-dev
```

[illegible]

ROOT - Download

Release 6.36.04 - 2025-08-25

Release Notes

The release notes for this release can be found [here](#).

Binary distributions

Instead of manually downloading this binary, please explore first whether your [package manager](#) already provides this version. This way, you will automatically keep up-to-date with the latest stable versions with no manual maintenance on your side.

Platform	Files	Size
Almalinux 8.10	root_v6.36.04.Linux-almalinux8.10-x86_64-gcc8.5.tar.gz	276M
Almalinux 9.6	root_v6.36.04.Linux-almalinux9.6-x86_64-gcc11.5.tar.gz	296M
Almalinux 10.0	root_v6.36.04.Linux-almalinux10.0-x86_64-gcc14.2.tar.gz	305M
Debian 12	root_v6.36.04.Linux-debian12-x86_64-gcc12.2.tar.gz	283M
Fedora 41	root_v6.36.04.Linux-fedora41-x86_64-gcc14.3.tar.gz	285M
Fedora 42	root_v6.36.04.Linux-fedora42-x86_64-gcc15.2.tar.gz	304M
Ubuntu 22.04	root_v6.36.04.Linux-ubuntu22.04-x86_64-gcc11.4.tar.gz	286M
Ubuntu 24.04	root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz	282M
Ubuntu 25.04	root_v6.36.04.Linux-ubuntu25.04-x86_64-gcc14.2.tar.gz	285M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.pkg	408M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.tar.gz	268M

ROOT - Download

Release 6.36.04 - 2025-08-25

Release Notes

The release notes for this release can be found [here](#).

Binary distributions

Instead of manually downloading this binary, please explore first whether your [package manager](#) already provides this version. This way, you will automatically keep up-to-date with the latest stable versions with no manual maintenance on your side.

Platform	Files	Size
Almalinux 8.10	root_v6.36.04.Linux-almalinux8.10-x86_64-gcc8.5.tar.gz	276M
Almalinux 9.6	root_v6.36.04.Linux-almalinux9.6-x86_64-gcc11.5.tar.gz	296M
Almalinux 10.0	root_v6.36.04.Linux-almalinux10.0-x86_64-gcc14.2.tar.gz	305M
Debian 12	root_v6.36.04.Linux-debian12-x86_64-gcc12.2.tar.gz	283M
Fedora 41	root_v6.36.04.Linux-fedora41-x86_64-gcc14.3.tar.gz	285M
Fedora 42	root_v6.36.04.Linux-fedora42-x86_64-gcc15.2.tar.gz	304M
Ubuntu 22.04	root_v6.36.04.Linux-ubuntu22.04-x86_64-gcc11.4.tar.gz	286M
Ubuntu 24.04	root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz	282M
Ubuntu 25.04	root_v6.36.04.Linux-ubuntu25.04-x86_64-gcc14.2.tar.gz	285M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.pkg	408M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.tar.gz	268M



Click tasto destro

Release 63604 - ROOT

root.cern/releases/release-63604/

Verifica la tua identità



ROOT
Data Analysis Framework

[About](#)[Install](#)[Manual](#)[Contribute](#)[For Developers](#)[Source](#)

Useful links

All releases

Release 6.36.04 - 2025-08-25

Release Notes

The release notes for this release can be found [here](#).

Binary distributions

Instead of manually downloading this binary, please explore first whether your [package manager](#) already provides this version. This way, you will automatically keep up-to-date with the latest stable versions with no manual maintenance on your side.

Platform	Files
Almalinux 8.10	root_v6.36.04.Lin
Almalinux 9.6	root_v6.36.04.Lin
Almalinux 10.0	root_v6.36.04.Lin
Debian 12	root_v6.36.04.Lin
Fedora 41	root_v6.36.04.Lin
Fedora 42	root_v6.36.04.Lin
Ubuntu 22.04	root_v6.36.04.Lin
Ubuntu 24.04	root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz 282M
Ubuntu 25.04	root_v6.36.04.Linux-ubuntu25.04-x86_64-gcc14.2.tar.gz 285M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.pkg 408M
macOS 13.7 arm64 Xcode 15	root_v6.36.04.macos-13.7-arm64-clang150.tar.gz 268M

On this page

[Release 6.36.04 - 2025-08-25](#)

[Release Notes](#)

[Binary distributions](#)

[Source distribution](#)

[Installations in CVMFS](#)

[Git](#)

[Windows](#)

[Important installation notes](#)

- Apri link in un'altra scheda
- Apri link in un'altra finestra
- Apri link in finestra di navigazione in incognito
- Salva link con nome...
- Copia indirizzo link
- Ispeziona

ROOT – Installazione

```
sg@LAPTOP-4KU4SPH9: ~/ROOT
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ mkdir ROOT
sg@LAPTOP-4KU4SPH9:~$ 
sg@LAPTOP-4KU4SPH9:~$ cd ROOT/
sg@LAPTOP-4KU4SPH9:~/ROOT$ 
sg@LAPTOP-4KU4SPH9:~/ROOT$ 
sg@LAPTOP-4KU4SPH9:~/ROOT$ wget https://root.cern/download/root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz
--2025-09-30 22:08:16-- https://root.cern/download/root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz
Resolving root.cern (root.cern)... 137.138.6.72, 2001:1458:201:8b::100:23b
Connecting to root.cern (root.cern)|137.138.6.72|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 296152492 (282M) [application/x-gzip]
Saving to: 'root_v6.36.04.Linux-ubuntu24.04-x86_64-gcc13.3.tar.gz'

6.04.Linux-ubuntu24.04-x86_64    0%[
```

ROOT – Installazione

```
sg@LAPTOP-4KU4SPH9: ~/ROOT
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$ mkdir ROOT
sg@LAPTOP-4KU4SPH9:~$
sg@LAPTOP-4KU4SPH9:~$ cd ROOT/
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ wget https://root.cern/download/root_v6.32.04.Linux-ubuntu24.04-x86_64-gcc13.2.tar.gz
--2024-08-26 15:49:50-- https://root.cern/download/root_v6.32.04.Linux-ubuntu24.04-x86_64-gcc13.2.tar.gz
Resolving root.cern (root.cern)... 188.184.49.144, 2001:1458:d00:55::100:c
Connecting to root.cern (root.cern)|188.184.49.144|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 298450707 (285M) [application/x-gzip]
Saving to: 'root_v6.32.04.Linux-ubuntu24.04-x86_64-gcc13.2.tar.gz'

root_v6.32.04.Linux-ubuntu24. 100%[=====>] 284.62M 7.58MB/s in 48s

2024-08-26 15:50:41 (5.96 MB/s) - 'root_v6.32.04.Linux-ubuntu24.04-x86_64-gcc13.2.tar.gz' saved [298450707/298450707]

sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ ls
root_v6.32.04.Linux-ubuntu24.04-x86_64-gcc13.2.tar.gz
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ tar -xvzf root_v6.32.04.Linux-ubuntu24.04-x86_64-gcc13.2.tar.gz
```

ROOT – Installazione

```
sg@LAPTOP-4KU4SPH9: ~/ROOT
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ ls -l root
total 152
-rw-r--r--  1 sg sg   989 Aug 14 05:35 LICENSE
drwxr-xr-x  3 sg sg  4096 Aug 14 06:01 README
drwxr-xr-x  2 sg sg  4096 Aug 14 06:01 bin
drwxr-xr-x  3 sg sg  4096 Aug 14 06:01 cmake
drwxr-xr-x  2 sg sg  4096 Aug 14 06:01 config
drwxr-xr-x  8 sg sg  4096 Aug 14 06:01 etc
drwxr-xr-x  2 sg sg  4096 Aug 14 06:01 fonts
drwxr-xr-x  3 sg sg  4096 Aug 14 06:01 geom
drwxr-xr-x  2 sg sg 20480 Aug 14 06:01 icons
drwxr-xr-x 13 sg sg 65536 Aug 14 06:01 include
drwxr-xr-x  7 sg sg  4096 Aug 14 06:01 js
drwxr-xr-x  9 sg sg 12288 Aug 26 15:53 lib
drwxr-xr-x  2 sg sg  4096 Aug 14 06:01 macros
drwxr-xr-x  3 sg sg  4096 Aug 14 06:01 man
drwxr-xr-x 49 sg sg  4096 Aug 14 06:01 tutorials
drwxr-xr-x 10 sg sg  4096 Aug 14 06:01 ui5
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
```

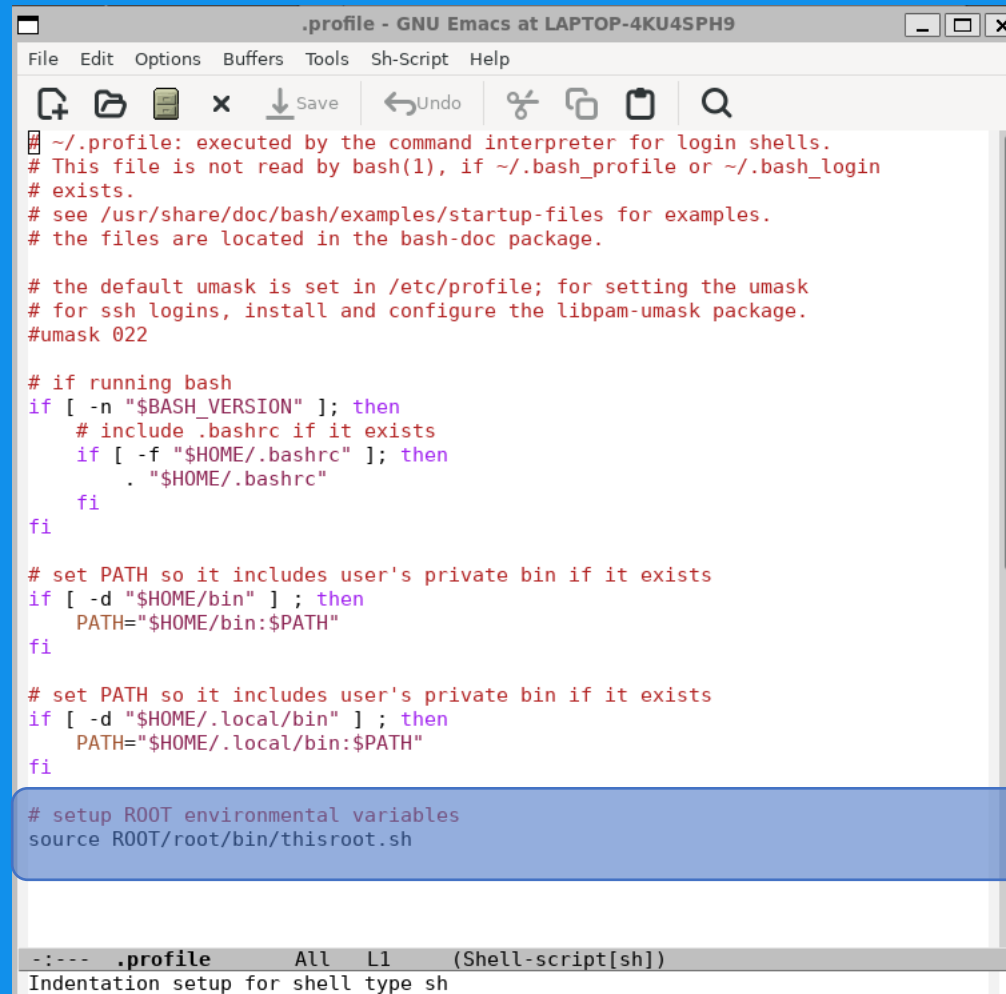
ROOT – controllo installazione

```
sg@LAPTOP-4KU4SPH9: ~/ROOT
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ source root/bin/thisroot.sh
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ root -l
root [0]
root [0]
root [0] .q
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$
sg@LAPTOP-4KU4SPH9:~/ROOT$ python3
Python 3.12.3 (main, Jul 31 2024, 17:43:48) [GCC 13.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>>
>>> import ROOT
>>>
>>>
sg@LAPTOP-4KU4SPH9:~/ROOT$
```

ROOT – attivazione automatica

```
sg@LAPTOP-4KU4SPH9: ~  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$  
sg@LAPTOP-4KU4SPH9:~/ROOT$ cd  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ ls -la  
total 64  
drwxr-x--- 8 sg sg 4096 Aug 26 15:57 .  
drwxr-xr-x 4 root root 4096 Aug 26 14:29 ..  
-rw----- 1 sg sg 1612 Aug 26 15:49 .bash_history  
-rw-r--r-- 1 sg sg 220 Aug 26 14:29 .bash_logout  
-rw-r--r-- 1 sg sg 3771 Aug 26 14:29 .bashrc  
drwx----- 4 sg sg 4096 Aug 26 15:06 .cache  
drwxr-xr-x 3 sg sg 4096 Aug 26 15:06 .config  
-rw-r--r-- 1 sg sg 515 Aug 26 14:41 .emacs  
drwx----- 4 sg sg 4096 Aug 26 14:41 .emacs.d  
drwxr-xr-x 2 sg sg 4096 Aug 26 14:29 .landscape  
drwxr-xr-x 4 sg sg 4096 Aug 26 15:05 .local  
-rw-r--r-- 1 sg sg 0 Aug 26 14:29 .motd_shown  
-rw-r--r-- 1 sg sg 878 Aug 26 15:33 .profile  
-rw----- 1 sg sg 540 Aug 26 15:54 .python_history  
-rw-r--r-- 1 sg sg 21 Aug 26 15:54 .root_hist  
-rw-r--r-- 1 sg sg 0 Aug 26 14:38 .sudo_as_admin_successful  
-rw-r--r-- 1 sg sg 164 Aug 26 15:50 .wget-hsts  
drwxr-xr-x 3 sg sg 4096 Aug 26 15:52 ROOT  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$  
sg@LAPTOP-4KU4SPH9:~$ emacs .profile &
```

ROOT – attivazione automatica



```
#!/.profile: executed by the command interpreter for login shells.
# This file is not read by bash(1), if ~/.bash_profile or ~/.bash_login
# exists.
# see /usr/share/doc/bash/examples/startup-files for examples.
# the files are located in the bash-doc package.

# the default umask is set in /etc/profile; for setting the umask
# for ssh logins, install and configure the libpam-umask package.
#umask 022

# if running bash
if [ -n "$BASH_VERSION" ]; then
    # include .bashrc if it exists
    if [ -f "$HOME/.bashrc" ]; then
        . "$HOME/.bashrc"
    fi
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/bin" ] ; then
    PATH="$HOME/bin:$PATH"
fi

# set PATH so it includes user's private bin if it exists
if [ -d "$HOME/.local/bin" ] ; then
    PATH="$HOME/.local/bin:$PATH"
fi

# setup ROOT environmental variables
source ROOT/root/bin/thisroot.sh
```

Righe da aggiungere

ROOT - Test

- Chiudere e riaprire Ubuntu (o aprire un altro terminale)
- Aprire il prompt python
- Eseguire i seguenti comandi:

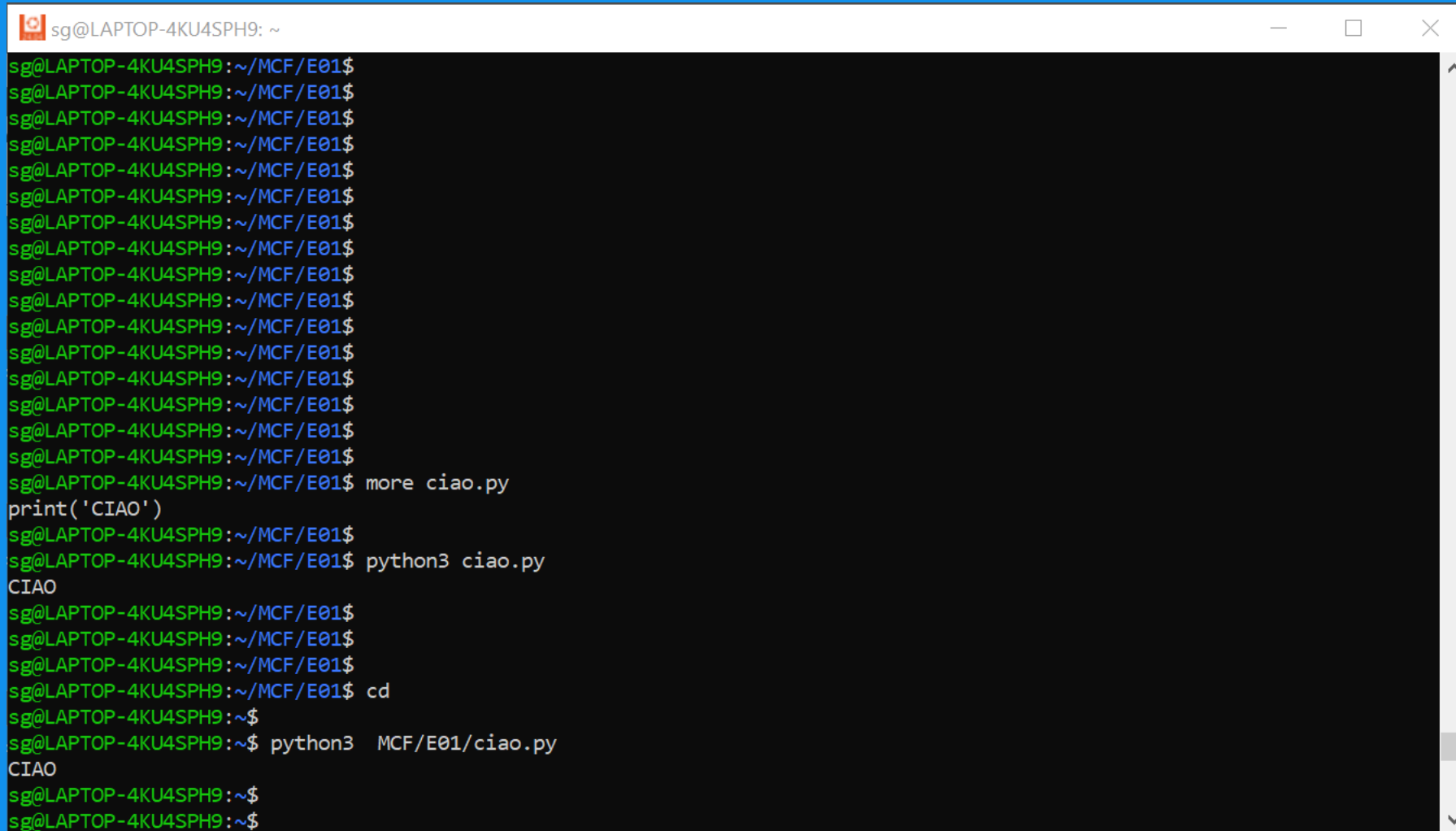
```
import ROOT
hh = ROOT.TH1F("hh", "My Histogram", 100, 0, 1)
hh.Draw()
```

```
sg@LAPTOP-4KU4SPH9:~$ python3
Python 3.12.3 (main, Jul 31 2024, 17:43:48) [GCC 13.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>>
>>> import ROOT
>>>
>>>
>>> hh = ROOT.TH1F("hh", "My Histogram", 100, 0, 1)
>>>
>>> hh.Draw()
Info in <TCanvas::MakeDefCanvas>:  created default TCanvas with name c1
>>>
```

Esercizio 2

1. Aprire il terminale Linux (Nativo o WSL);
2. Spostarsi all'interno della cartella *MCF* precedentemente creata;
3. Creare la sottocartella *E01* e spostarsi al suo interno;
4. Creare il file di testo *ciao.py* (script python);
5. Editare il file inserendo la seguente istruzione:
`print('CIAO!')`
6. Assicurarsi di aver salvato correttamente il file
 - Eventualmente provare il comando:
`less ciao.py`
7. Eseguire lo script python nel prompt dei comandi attraverso:
`python3 ciao.py`
8. Spostarsi nella cartella *home* ed eseguire lo script tramite il comando;
`python3 MCF/E01/ciao.py`

Esercizio 2



```
sg@LAPTOP-4KU4SPH9: ~
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$ more ciao.py
print('CIAO')
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$ python3 ciao.py
CIAO
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$
sg@LAPTOP-4KU4SPH9: ~/MCF/E01$ cd
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$ python3 MCF/E01/ciao.py
CIAO
sg@LAPTOP-4KU4SPH9: ~$
sg@LAPTOP-4KU4SPH9: ~$
```